

Operation and Maintenance Manual

Diesel engine-generator sets
with MTU 10/12 V 1600 Gx0 engines
Control version 1
Built in Europe

SS180008/02E



System designation	Engine model	Application group
DP00450D5S	10V1600G10F	3B, Continuous operation, variable
DP00500D5S	10V1600G20F	3B, Continuous operation, variable
DS00500D5S	10V1600G70F	3D, Short-term operation, fuel stop
DS00550D5S	10V1600G80F	3D, Short-term operation, fuel stop
DP00590D5S	12V1600G10F	3B, Continuous operation, variable
DP00650D5S	12V1600G20F	3B, Continuous operation, variable
DS00650D5S	12V1600G70F	3D, Short-term operation, fuel stop
DS00715D5S	12V1600G80F	3D, Short-term operation, fuel stop

Table 1: Applicability

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Preface

This Operation & Maintenance Manual provides general instructions for operating your MTU Onsite Energy engine-generator set properly. It is essential that every person who works on or with the engine-generator set be completely familiar with the contents of this manual and that he/she carefully follows the instructions contained herein.

Each operation and maintenance may require some modification of the suggested guidelines in this manual. They must be consistent with locally applicable standards and take into consideration safety guidelines and measures.

Following this guide will result in an efficient and reliable installation.

Carefully follow all procedures and safety precautions to ensure proper equipment operation and to avoid bodily injury. Read and follow the Safety section at the beginning of this manual.

All instructions and diagrams have been checked for accuracy and simplicity of application, however, the skills of the operator are most important. MTU Onsite Energy does not guarantee the result of any operation contained in this manual nor can MTU Onsite Energy assume responsibility for any injury or damage to property. Persons engaging in operating do so entirely at their own risk.

In this publication the manufacturer of the engine-generator set is known as MTU Onsite Energy.

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1 Safety

1.1 Important provisions for all products

Nameplate

The product is identified by nameplate, model designation or serial number and must match with the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Noncompliance with the safety instructions and warning notices

Correct use

The product is intended exclusively for the application specified in the contract or defined at the time of delivery.

This means that the equipment must be operated:

- Within the permissible operating parameters in accordance with the (→ product data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ applicable Spare Parts Catalog)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine control/parameters)
- In compliance with all safety instructions and in adherence to all warning notices in this manual
- In accordance with the maintenance requirements over the entire service life of the product (→ Maintenance Schedule)
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use is considered improper use and increases the risk of personnel injury or material damage in product operation. The manufacturer will accept no liability for such damage.

Modifications or conversions

Unauthorized modifications to the product compromise safety.

The manufacturer will accept no liability or warranty claims for any damage caused by unauthorized modifications or conversions.

Spare parts

Only genuine spare parts must be used to replace components or assemblies.

The manufacturer will accept no liability or warranty claims for any damage caused by the use of other spare parts.

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Transport

Lifting instruction

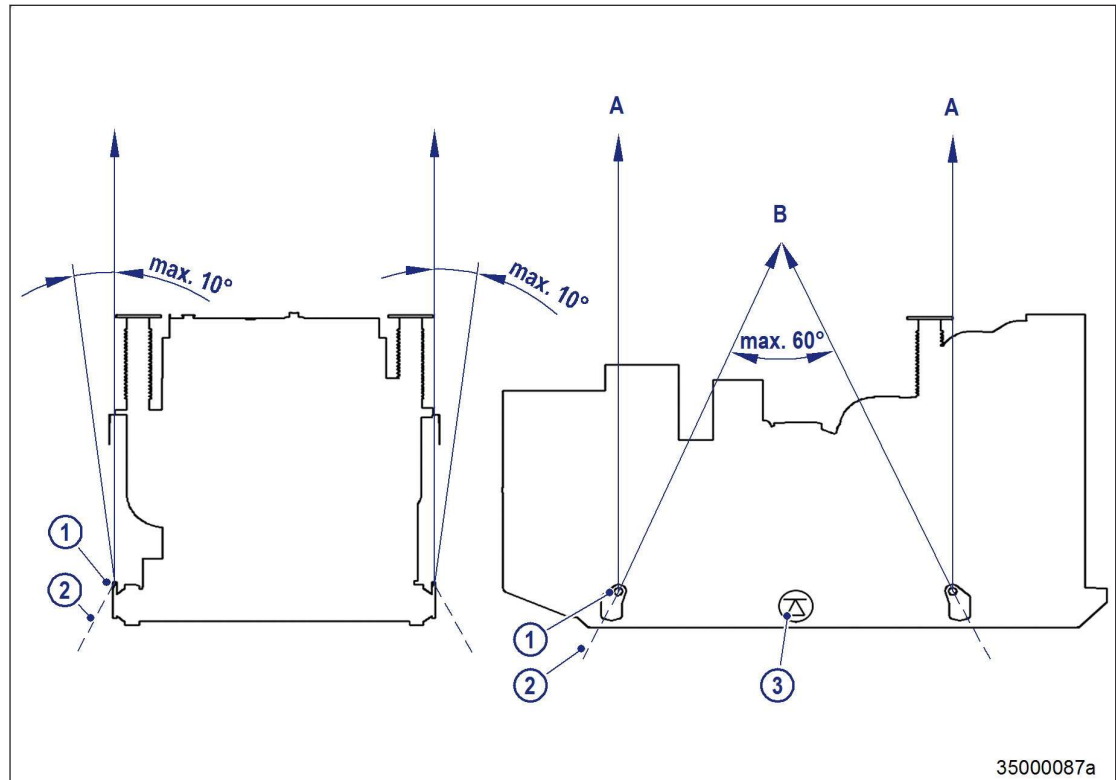


Figure 1: Lifting bracket location and recommended lifting procedures for engine-generator set with 10/12 V 1600 engine

- | | | |
|------------------------|-------------------|---------------------|
| A H cross beam variant | 1 Lifting bracket | 3 Center of gravity |
| B Cross beam variant | 2 Stabilizer rope | |

The (→ Figure 1) shows the lifting instructions for engine-generator sets:

- Variant A – vertical lifting with an H-cross beam.
- Variant B – diagonal lifting with cross beam. The resulting angle must not exceed 60°.

The engine-generator set should only be lifted by the lifting brackets (1) on the base frame.

Do not use engine or generator lifting eyes to lift the engine-generator set.

Pay attention to the center of gravity (3) of the engine-generator set.

To stabilize the load, attach stabilizer ropes (2) on the lifting eyes.

Only use transport and lifting devices approved by MTU.

Remove any loose parts on the engine-generator set.

Raise the engine-generator set slowly. The lifting ropes or chains must not make contact with the engine-generator set.

Transport

Fit engine mount locking devices and generator mount locking devices prior to transport.

The engine-generator set must only be transported in installation position: maximum permissible diagonal pull 10°.

Secure the engine-generator set such as to preclude tipping during transport. Secure such as to preclude slipping and tipping when driving up or down inclines and ramps.

1.4 Safety regulations for startup and operation

Safety regulations for startup

Install the product correctly and carry out acceptance in accordance the manufacturer's specifications before putting the product into service.

Before the product is put into operation for the first time, all official authorizations must be available and commissioning preconditions met.

When putting the product into operation, always ensure

- that all maintenance and repair work has been completed;
- that all loose parts have been removed from rotating machine components;
- that no-one is present in the danger zone of rotating machine components.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the signaling and alarm systems work properly.

Safety regulations for equipment operation

The operator must be familiar with the control and display elements.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be permanently observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

The procedures for cases of emergency, in particular, emergency stop, must be practiced regularly.

The following steps must be taken if a malfunction of the system is recognized or reported by the system:

- Inform supervisor(s) in charge,
- Analyze the message,
- If required, carry out emergency operations e.g. emergency stop.

Operation

The following conditions must be fulfilled before starting the product:

- Wear ear protection.
- Ensure that the engine room is well ventilated.
- Do not inhale the exhaust gases of the product.
- Ensure that the exhaust system is free of leaks and that the gases are discharged to atmosphere.
- Mop up any leaked or spilt fluids and lubricants immediately or soak up with a suitable binding agent.
- Protect battery terminals, generator terminals or cables against accidental contact.

Operation of electrical equipment

When electrical equipment is in operation, certain components of these appliances are electrically live.

Observe the safety instructions for these devices.